Gender Hiring Quotas

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Abstract

Gender quotas have mostly targeted corporate boards and done little for the gender employment gap. Requiring a minimum share of women in a firm's entire workforce would force firms to lay off workers or hire beyond their needs. A quota on new hires avoids these costs, but firms can comply and then undo its effect through separations. We study a novel type of quota: Italian firms that win public contracts must reserve for women at least 30 percent of the new hires needed for the project. Coverage depends on how contracts are funded, so similar contracts are awarded with and without the quota. Linking procurement records to employer-employee data and balance sheets, we compare winners of contracts with and without the quota in a matched difference-in-differences design. Three years after the award, female employment in firms subject to the quota is 11 percent higher than in comparable winners without it, with no difference for men, although some firms avoid the requirement by hiring less. Female separations rise once the contract ends, but do not fully offset the additional hires. The new jobs are mostly white-collar but temporary and part-time, with no effect on wages. We find no evidence of efficiency losses. Hiring quotas can bring women into male-dominated firms, but their effects partly fade once the obligation expires.

Keywords: Gender hiring quotas; Public procurement; Affirmative action; Labor demand; Female employment; Matched employer-employee data.

JEL Codes: H57, J16, J23, J78.

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1 Introduction

Despite decades of progress, gender gaps persist both in employment and in top positions (Olivetti and Petrongolo, 2016; Blau and Kahn, 2017). In OECD countries, 72.5 percent of women are employed, against 87.7 percent of men.[1] To close the gap at the top, many countries have introduced gender quotas for corporate boards.[2] These quotas have raised women's presence on boards, but they reach a few, highly selected women. They can narrow the employment gap only indirectly, through spillovers to women below the top, and studies find little such effect (Bertrand et al., 2019; Maida and Weber, 2022). Acting on the employment gap directly through a quota on the stock of employment would force firms either to break existing matches or to hire beyond their labor demand. A quota on new hires avoids both problems: it can nudge firms toward a more gender-balanced workforce as they replace and expand their staff. Whether it works depends on three questions. First, is the effect long lasting, or do firms undo it through separations? Second, do firms comply but relegate the new women to lower positions? Third, since firms were hiring few women before, does the quota create large efficiency losses?

We answer these questions using a gender hiring quota introduced in Italy, which has one of the largest gender employment gaps in the EU.[3] Since 2021, firms that win public contracts must reserve for women at least 30 percent of the new hires needed for the project.[4] The setting is ideal for three reasons. First, the quota binds: most of the firms we study operate in construction, and before the award women account for only 9 percent of their hires, less than a third of the share the quota requires. Second, firms have strong incentives to comply, because the contract is worth on average 32 percent of their annual sales.[5] Third, the quota expires with the contract, so we can test whether its effects last. We find that the quota raises female employment at no cost to firm performance. Most new jobs, however, are temporary white-collar positions, and part of the gain fades once the contract ends.

The quota applies to public contracts funded by the National Recovery and Resilience Plan (PNRR).[6] Contracting authorities specify it ex ante in the call for tenders, so firms

[1] Employment rates refer to the population aged 25-54 (OECD, 2024).

[2] Norway introduced a 40 percent board quota in 2003. Today 28 jurisdictions have binding board quotas, and Directive (EU) 2022/2381 extends them to large listed firms across the EU.

[3] In 2024, the employment rate of women aged 25-54 was 19.2 percentage points below that of men in Italy, twice the EU average of 9.6 points (Eurostat, 2024).

[4] Existing affirmative-action programs typically set non-binding goals or target narrow groups of workers (Leonard, 1984; Miller, 2017; Szerman, 2022).

[5] Computed for the 87.4 percent of winners for which we observe balance sheets. The median is 5.5 percent.

[6] The PNRR is Italy's component of Next Generation EU, the EU's post-COVID-19 recovery program, running from 2021 to 2026. PNRR contracts accounted for 8.1 percent of the value of public procurement in 2021-2025, peaking at about 18 percent in 2023. The quota is set by Article 47 of Decree-Law 77/2021,

know whether a contract carries it before deciding to bid.[7] Waivers are hard to obtain: contracting authorities must formally justify why the quota is infeasible, for instance because the contract is small or requires fewer than three hires. By contrast, waivers are generally not accepted for labor-intensive contracts.[8] In sectors with particularly low female employment, contracting authorities may reduce the quota target, but not waive it entirely. In practice, the quota applies in full to about 42 percent of covered contracts, is reduced in 5 percent, and is waived in about 53 percent. After the award, the quota binds only if the firm hires at least three workers for the project, and leaves firms free to choose how many workers to hire and in which occupations, hours, and contract types. Firms can therefore hire more women, hire fewer men, or stay below the three-hire threshold to avoid the requirement.

We build a unique dataset that links employer-employee records to the universe of Italian public procurement contracts and to firm balance sheets. Because coverage depends mainly on how a contract is funded, not on what it buys, we can compare firms awarded contracts with the quota to firms awarded comparable contracts without it, using a matched difference-in-differences design. Both groups win a public contract, so the comparison isolates the effect of the quota from that of winning. We match contracts on type of work, procedure, timing, and value, and firms on sector, size, and pre-award female employment share.[9] The key assumption is that, without the quota, two similar firms winning similar contracts would have hired and retained women in the same way.

Covered contracts are important for the firms that win them: a firm is unlikely to give up a large public contract to avoid hiring a woman. Selection into the quota is therefore likely to be a second-order concern, even though firms know before bidding whether a contract carries it. Matching on pre-award female employment and the absence of pre-trends further address this concern.

The firms in our sample are small, male-dominated, and hire infrequently. In the year before the award, the median firm employs 12 workers, one of whom is a woman, and hires three new workers, none of them women. For 46 percent of firms, complying would mean hiring a woman for the first time. Women employed before the award hold mostly white-collar positions. Seventy percent of the firms operate in construction, where only 8 percent

which also sets an independent 30 percent quota for workers under 36.

[7]Contracting authorities are the public entities that buy works, goods, and services, from ministries and regions to municipalities, health authorities, and publicly owned companies. They design the call for tenders, choose the award procedure, and select the winning firm.

[8]The implementing guidelines call this a "particularly stringent" burden, to be formalized before the procedure starts (Prime Minister's Decree of 7 December 2021). Labor-intensive contracts are those in which labor costs account for at least half of the contract value.

[9]The unit of analysis is the firm-contract, and treatment is a firm's first award of a contract with the quota. Before the award, treated and control firms employ on average 20.7 and 20.9 workers, and their contracts have nearly identical values.

of workers are women[10] and physical strength gives men a productivity advantage (Galor and Weil, 1996; Pitt et al., 2012). Our setting is therefore a demanding test of the quota: if it works here, it is likely to work where women are easier to hire.

The quota raises female employment by 11.2 percent three years after the award, about 0.3 additional women per firm, with no change in male employment. The increase is driven by female hiring, while the probability of any hire falls by 2.8 percentage points, reflecting fewer male hires and consistent with some firms avoiding the requirement. First, the gain partly fades: female separations do not rise while the contract is in force, but increase once it ends, without fully offsetting the additional hires. Second, most additional women come from non-employment and are hired into white-collar jobs, but 81 percent of these jobs are temporary and 64 percent part-time, with no effect on wages. Third, we find no evidence of efficiency losses: revenues, assets, and profit margins do not change, and labor productivity shows a small positive effect.

Two forces shape these responses. The first is the government's leverage as a buyer. Responses are stronger when the contract accounts for a larger share of firm sales, consistent with firms complying because they depend on public demand. The second is the combination of temporary contracts and hiring frictions, which explains why the gain fades but does not disappear. Temporary contracts can simply expire, so separations rise once the contract ends. Yet replacing a woman with a man requires a firing cost and a new hiring cost, and she accumulates skills specific to the firm. The quota thus creates a wedge between keeping the new hires and returning to the old workforce composition, even after the quota expires. Even when the jobs end, they may strengthen women's attachment to the labor market: most of these women come from non-employment, and work experience can make it easier to find the next job.

Contribution to the Literature Our paper contributes to three literatures. First, we add to the literature on gender quotas. So far, quotas have targeted positions of power, in politics and on corporate boards. They can change beliefs about women in leadership (Beaman et al., 2009, 2012) and the quality of those selected (Baltrunaite et al., 2014). In firms, evidence on performance is mixed, with some studies finding costs (Ahern and Dittmar, 2012) and others no effect (Eckbo et al., 2022; Ferrari et al., 2022), and board quotas have little effect on women below the top (Bertrand et al., 2019; Maida and Weber, 2022). Whether quotas can close the employment gap therefore remains an open question. We study a quota that acts directly on hiring, across all jobs, and examine the questions

[10]In 2024, women accounted for 8.5 percent of workers aged 25-64 employed in construction in Italy (Eurostat, 2024).

this raises: whether its effects last, which jobs women get, and whether firms bear efficiency costs.

Second, we contribute to the literature on affirmative action in employment. Existing programs fall into two groups. The first sets hiring goals for broad groups, as in the US federal contractor program, which ties affirmative action to public contracts but does not bind (Leonard, 1984; Miller, 2017; Amano-Patino et al., 2021). The second binds, but applies to the stock of employment and to narrow groups, such as persons with disabilities (Szerman, 2022). We study a quota that is binding, applies to the flow of new hires, and targets half of the population. Because it expires with the contract, we can also measure how much of the response survives once the constraint is lifted.

Finally, we contribute to the literature on public procurement as a policy tool. Governments often use their purchasing power to favor particular suppliers, and existing work studies how these preferences affect who wins contracts, procurement costs, and competition (Marion, 2007, 2009; Krasnokutskaya and Seim, 2011; Athey et al., 2013; Rosa, 2024). Conditions on what winning firms do after the award have received less attention, and the main example, the US federal contractor program, sets non-binding goals. We show that a binding condition attached to public contracts can change the hiring of private employers, and that its effect grows with the firm's reliance on public demand.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting, Section 3 the data, and Section 4 the empirical strategy. Section 5 presents a simple framework for how firms can respond to the quota and the results: the effects on female employment, the margins along which firms adjust, firm performance, and heterogeneity by the economic importance of the contract and by firm size. Section 6 concludes.

2 Institutional Setting

2.1 Public Procurement in Italy

Public procurement is one of the largest channels through which governments buy from private firms. Across OECD countries, it accounts for 12.7 percent of GDP (OECD, 2023). Italy is close to this average: public bodies spend more than €300 billion every year on works, services, and supplies.

The Italian procurement system is highly decentralized. Public calls are issued and managed by contracting authorities. There are tens of thousands of them, including ministries, regions, municipalities, health agencies, and public companies. Each authority defines the object of the contract, chooses the award procedure, sets the reserve price, and selects the

winner, following the rules of the Public Contracts Code.[11]

Each contract moves funds from the government to a private firm through a contracting authority. The government allocates funds to the contracting authority. The authority issues the call, selects the winning firm, and signs the contract. The firm then executes the project, bears all contractual obligations, and receives payment.

Firms can bid for a contract alone or as part of a group. When firms bid together, they form a temporary consortium and appoint a lead firm. The lead firm signs the contract on behalf of all members and represents them before the contracting authority. All members remain jointly liable for the contractual obligations.

Each contract binds the firm for a limited period. The contract sets the time within which the firm must complete the project. All obligations attached to the contract, including those on hiring, apply only over this period.

The contracting authority monitors how the contract is executed. It appoints a project manager, who is responsible for the procedure from the call to the final payment. It also appoints a works director, or a contract supervisor for services and supplies, who checks that the firm performs as agreed. At the end of the contract, a final inspection certifies that the work conforms to the contract before the firm receives its last payment.[12]

Firms that do not comply with the contract face financial penalties and, in serious cases, termination. Every contract must set daily penalties for delays, between 0.05 and 0.15 percent of the net contract value per day, up to a total of 10 percent of that value.[13] When penalties exceed this cap, or when the firm commits a serious breach, the authority can terminate the contract. Before signing, the firm must also post a performance guarantee, typically equal to 10 percent of the contract value, which the authority can draw on if the firm fails to perform.[14] Serious breaches are recorded in a registry kept by ANAC and can exclude the firm from future calls. For firms that depend on public demand, non-compliance can therefore be costly beyond the single contract.

Winning a contract creates a labor demand shock, and the firm can meet it in two ways. First, it can decide not to hire new workers and simply reallocate its existing employees to the project. Second, it can hire the new workers needed to execute the project, on top of its existing workforce. This choice matters for our setting because, as we explain below, the gender quota applies only to new hires.

[11] Legislative Decree 50/2016, replaced by Legislative Decree 36/2023 from July 2023.

[12] The project manager is the *Responsabile unico del progetto* (RUP), the works director is the *direttore dei lavori*, and the contract supervisor is the *direttore dell'esecuzione*. The final inspection is the *collaudo* for works and the *verifica di conformità* for services and supplies.

[13] Article 126 of Legislative Decree 36/2023. Under Legislative Decree 50/2016 (Article 113-bis), the daily range was 0.03 to 0.1 percent, with the same 10 percent cap.

[14] Article 117 of Legislative Decree 36/2023.

2.2 The PNRR and the Gender Hiring Quota

The National Recovery and Resilience Plan (PNRR) is Italy's share of NextGenerationEU, the program the European Union launched after the COVID-19 pandemic to repair its immediate economic and social damage.[15] Italy is the program's largest recipient. Between 2021 and 2026, the plan channels €191.5 billion into the Italian economy, mostly through public procurement. Because the funds must be spent by 2026, the plan generated a large and concentrated wave of public contracts on top of ordinary procurement. PNRR contracts accounted for 8.1 percent of the value of Italian public procurement between 2021 and 2025, and for about 18 percent in 2023. In our data, PNRR funds finance 59,726 contracts, worth €22.7 billion. Of these, 20,367 contracts, worth €10.6 billion, carry the full gender quota.

Article 47 of Decree-Law 77/2021 attached a hiring requirement to these contracts, to align PNRR spending with the plan's objectives on gender and generational equality.[16] Firms that win a PNRR-funded contract must reserve for women at least 30 percent of the new hires needed to execute it. A separate 30 percent requirement applies to workers younger than 36, and the two targets must be met independently. The requirement also extends to subcontractors working on the same contract. In this paper we focus on the gender requirement.

The quota targets the flow of new hires, not the firm's existing workforce. It covers all hires on employment contracts needed to execute the project, at any level of the firm. The policy only sets the share of women among new hires and leaves firms full discretion over the jobs they offer. It does not prescribe occupations, wages, working hours, or contract types. A firm can therefore meet the quota by hiring women into blue-collar or white-collar positions, on permanent or temporary contracts, and full-time or part-time. Like all contractual obligations, the quota applies only for the duration of the contract.

Contracting authorities decide the quota status of each contract before the call is published. They can apply the full 30 percent quota, reduce it, or waive it, but any deviation requires an "adequate and specific justification." The implementing guidelines describe this as a "particularly stringent" burden. The justification must be formalized in an act signed by the head of the contracting authority before or at the start of the procedure, typically the decision to contract (*determina a contrarre*), and must show that the quota is infeasible or conflicts with the efficiency, cost-effectiveness, or quality of the service. The guidelines give four examples of full waivers: direct awards of small value; procedures expected to re-

[15] NextGenerationEU is a temporary instrument financed through joint EU borrowing on capital markets.

[16] Decree-Law 77/2021 of May 31, 2021, converted into Law 108/2021. Implementing guidelines were adopted by Prime Ministerial Decree of December 7, 2021. The same rules apply to contracts financed by the National Complementary Plan (PNC).

quire fewer than three new hires; hires that require professional qualifications held by few workers in the short run; and procedures of extreme urgency or civil protection. Conversely, waivers are considered hard to justify for labor-intensive service contracts, in which labor costs account for at least half of the contract value. A waiver can instead be justified when social clauses require the winning firm to re-absorb the workers of the previous contractor. A partial reduction is allowed in sectors where female employment rates are far below the national average, but the quota cannot be waived entirely on these grounds. In that case the authority sets a lower target, which must still raise female hiring above the sectoral benchmark. In our data, the quota is waived in about 53 percent of PNRR contracts and reduced in 5 percent. Firms observe the quota status before deciding whether to bid. Once a contract with the quota is awarded, the firm must comply only if it hires at least three workers for the project.

The quota comes with possible penalties for non-compliance and with reporting duties for the winning firm. Each contract must set monetary penalties, proportional to the severity of the violation and to the value of the contract. Firms with 15 to 50 employees must also deliver a report on the gender composition of their workforce within six months of signing the contract. Failure to deliver this report prevents the firm from taking part in subsequent PNRR-funded tenders for twelve months. For firms that rely on public procurement, the expected cost of ignoring the quota may therefore be large.

In practice, enforcement of the quota is uncertain. The guidelines do not clearly designate the body in charge of verifying compliance, and we are not aware of public evidence on how often the quota was met or penalties were applied. We therefore observe whether a firm was exposed to the quota, not whether it complied with it. Our estimates measure the effect of the quota as it was monitored and enforced in practice, and we cannot separate low compliance from weak monitoring.

2.3 A Hiring Quota for the Entire Workforce

The Italian quota differs from the gender quotas studied so far in one central respect: it acts on the hiring of ordinary workers. Board quotas, such as those introduced in Norway and Italy, reserve a few seats at the top of large firms (Bertrand et al., 2019; Maida and Weber, 2022). Political quotas reserve elected positions (Beaman et al., 2009; Baltrunaite et al., 2014). Both types of quotas target a small number of positions. Their effect on the typical female worker depends on a trickle-down from women in leadership to women in the rest of the organization, and the evidence on this channel is mixed (Bertrand et al., 2019). The policy we study removes the need for any trickle-down. It applies to every new hire

the contract requires and acts on the flow of workers rather than on the stock. It therefore targets the recruitment of women, not their appointment to a few top positions.

The quota also differs from employment affirmative action as it is usually designed. In the United States, federal contractors must adopt affirmative action plans, but they face no binding numerical requirement on new hires (Leonard, 1984; Miller, 2017). Hiring quotas for workers with disabilities do set a numerical target, but they reserve a small share of jobs for a small group of workers (Szerman, 2022). Preference programs in public procurement, in turn, favor certain types of suppliers at the bidding stage, such as small or minority-owned firms (Marion, 2007, 2009; Krasnokutskaya and Seim, 2011; Athey et al., 2013). Our gender hiring quota does none of these. It sets a binding numerical target for women, who make up half of the potential workforce, and it applies to the hiring decisions of the winning firm rather than to the selection of the supplier.

Three features of the policy make it well suited to study how firms respond to a hiring quota. First, the quota binds exactly when firms expand. A contract award is a discrete and observable labor demand shock, and the quota constrains the hires that this shock generates. Second, quota status varies across contracts with the same economic content, for two reasons. Within the PNRR, contracting authorities apply, reduce, or waive the quota for similar works and services, in the same period and under the same procedures. Outside the PNRR, contracts with the same object are financed with ordinary funds and carry no quota at all. Both sources of variation let us compare firms that win similar contracts with and without the quota. Third, the quota is temporary. It lasts only as long as the contract, so we can observe what happens to female employment once the requirement expires.

Italy is also a setting where a hiring quota can matter most. In 2024, the employment rate of women aged 25-54 was 64.9 percent in Italy, against 77.6 percent in the EU. The gender employment gap was 19.2 percentage points, twice the EU average of 9.6 points and one of the largest in the European Union (Eurostat, 2024). In the firms that win PNRR contracts, the gap is even wider: women account for about 11 percent of employees and about 10 percent of new hires. A 30 percent requirement on new hires is therefore far from what these firms would choose on their own. A large pool of women outside employment also means that firms can, in principle, meet the quota without poaching female workers from other employers.

2.4 Where the Quota Binds

The quota falls mostly on sectors with very few women. Figure 1 reports, for each sector, its share of the 4,112 PNRR contracts subject to the quota and the distribution of the female

employment share among the firms that win them. Construction accounts for 59.5 percent of these contracts. Multi-sector firms account for 12.8 percent, manufacturing for 11.7 percent, and trade for 5.9 percent. ICT, professional services, administrative services, and utilities together account for less than 10 percent.

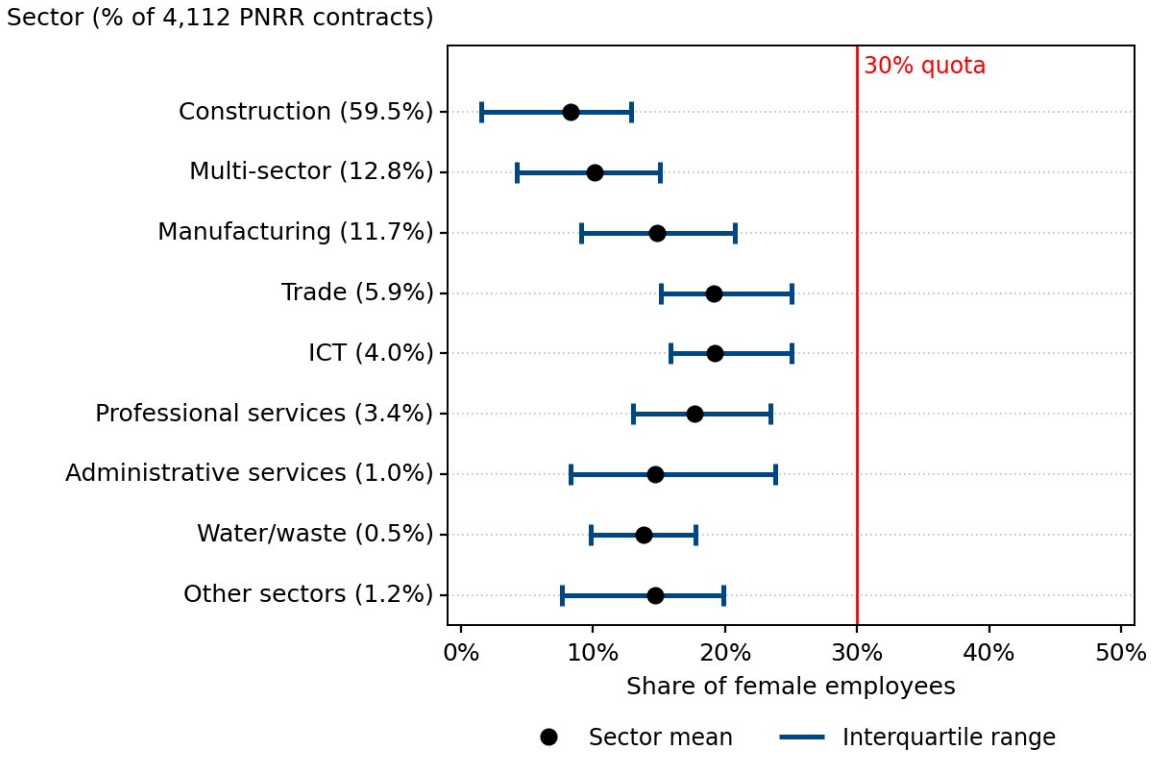


Figure 1: Exposure to the Quota by Sector

Notes: For each sector, the figure reports its share of the 4,112 PNRR contracts subject to the full quota and the distribution of the female employment share among the firms that win them.

In every sector, the typical winning firm employs far fewer women than the quota requires. The average female employment share is below 20 percent in all sectors and is lowest in construction, at about 8 percent. Even firms at the 75th percentile of each sector remain below the 30 percent threshold. The quota is therefore binding for almost every firm that wins a PNRR contract, and most binding where most contracts go.

Construction is central to our sample. Construction firms make up 71 percent of all treated firms and 73 percent of the matched treated firms. For this reason, our estimates speak mostly to how construction firms respond to a hiring quota. Construction relies on physically demanding tasks, where physical strength gives men a productivity advantage (Galor and Weil, 1996; Pitt et al., 2012; Rendall, 2017). Construction is therefore a demand-

ing test of the quota: if the quota raises female employment without efficiency losses here, it is likely to do so where women are easier to hire. The supply of women trained for blue-collar construction jobs is small. Meeting the quota is therefore costly, and firms have stronger incentives to avoid hiring or to place women in a narrow set of jobs. In sectors where the production technology does not favor men, firms can meet the quota at lower cost, and we expect larger employment responses. Of course, a low baseline share also leaves more room for female employment to grow. We return to this point in Section 5.6, where we compare responses across sectors.

3 Data: Linking Contracts to Workers

Our data link every public contract awarded in Italy from 2017 to 2025 to the full workforce of the firm that wins it. We combine the universe of public calls and their winners with matched employer-employee records covering all private-sector jobs, and with firm balance sheets. The three sources share firms' tax identifiers, so we can follow each winning firm and each of its workers before and after the award. For every PNRR call, we also observe whether the contracting authority applied, reduced, or waived the gender quota. This combination is rare. It lets us observe the exact contract that imposes the quota, all the hires the firm makes after the award, and the characteristics of every worker it recruits: gender, previous labor market status, occupation, contract, and wage. We can therefore measure each margin through which firms may respond to the quota and compare these responses with those of similar firms that win similar contracts without it.

3.1 Firm-Level Administrative Data

Our firm-level data combine matched employer-employee records, worker and firm demographics, and firm balance sheets. Together, they give us a full overview of who works in each firm, whom the firm hires, and how it performs.

The matched employer-employee data from the Italian Social Security Institute (INPS) contain every employment contract in the Italian private sector from 2017 to 2025. They are based on the monthly contribution declarations that firms file for each employee (UniEmens). For each contract, we observe the weekly wage, weeks worked, working time (full-time or part-time), contract type (permanent or temporary), and occupation (blue-collar or white-collar).

We use these contracts to build the outcomes that measure each margin of response to the quota. At the firm level, we construct total and female employment, the number of new

hires by gender, the female share of employment and of new hires, and separations. We measure employment as the number of workers with an employment contract with the firm at any point during the year, so it includes workers on short contracts. We define a new hire as a worker who is employed by the firm in a given year but not in the previous year; workers who return after at least one year away therefore count as new hires. A separation occurs when a worker employed by the firm in one year is no longer employed there in the next. Our data do not link workers to individual procurement contracts. We therefore cannot distinguish the hires needed to execute the project from the hires the firm would have made anyway, and we measure all new hires after the award. Because we follow workers across employers, we also observe where new hires come from. A new hire comes from another firm if the worker had a contract with a different private employer in the previous year, and from non-employment if the worker had no employment contract in our data in that year. Non-employment therefore includes unemployment and inactivity, but also jobs not covered by INPS records, such as self-employment and public-sector employment.

INPS also provides demographic information on workers and firms. For each worker, we observe age and gender. For each firm, we observe location, sector, and size. We use these characteristics to describe the firms that win public contracts and to match treated and control firms before the award.

Balance-sheet data from Orbis measure firm performance. For each firm, we observe sales, input costs, value added, profits, and total assets. We measure labor productivity as the log of value added per employee, as reported in Orbis.

3.2 Public Procurement Data

Our procurement data combine the universe of Italian public calls, the firms that win them, and the quota status of each PNRR call. Together, they tell us which contracts were awarded, to whom, and under which hiring requirement.

The records of the National Anti-Corruption Authority (ANAC) cover every call issued by an Italian contracting authority from 2017 to 2025. For each call, we observe the project description, the object of the contract (works, services, or supplies) and its Common Procurement Vocabulary (CPV) code, the award procedure, the location, the contract amount, the contracting authority, and the source of funding.[17] The CPV code classifies the object of each contract at a fine level, which lets us compare contracts with the same economic

[17]The CPV is the classification system the European Union uses to describe the object of public contracts. Each code has eight digits, plus a check digit, and is hierarchical: the first two digits identify a division, and each additional digit identifies a narrower group, class, and category. For example, division 30 covers office and computing machinery, while code 30197221 identifies paperclip holders.

content. ANAC also records the winners of each call over the same period. For each winning firm, we observe its role in the contract (sole winner, lead firm, or member of a temporary consortium) and the winning bid.

The quota status of PNRR calls comes from ReGiS, the monitoring system of the Italian Ministry of Economy and Finance, and covers calls issued from 2021 to 2024. For each call, we observe the level of the quota and whether the contracting authority waived or reduced it. We link ReGiS to ANAC through the tender identifier.

3.3 Sample Restrictions

We focus on public contracts awarded between 2021 and 2024, the years in which PNRR calls and their quota status are observed. We keep contracts worth between €15,000 and €10 million. [18] The lower bound excludes very small purchases, which are typically exempt from the quota. The upper bound excludes a small number of very large contracts for which comparable contracts without the quota are rare. We keep only private firms with at least one employee in the baseline year, defined as the year before the award. When a temporary consortium wins a contract, we keep only the lead firm. Our unit of analysis is the firm-contract, that is, a firm winning a given contract. The final sample contains 214,073 firm-contracts, won by 45,593 distinct firms.

3.4 Descriptive Statistics

ALL THIS SECTION IS GONNA BE COMPLETELY UPDATED AFTER FINISHING SOME DESCRIPTIVE ANALYSIS IN INPS

PNRR contracts are a large share of public procurement in our sample period. Table **??** compares PNRR contracts with ordinary contracts awarded in the same years. PNRR contracts account for XX percent of all contracts and XX percent of their total value. The median PNRR contract is worth €XX, against €XX for ordinary contracts. Works account for XX percent of PNRR contracts and XX percent of ordinary ones. PNRR contracts are also [shorter/longer]: the median contract lasts XX months, against XX months for ordinary contracts, and XX percent last less than one year.[19] The quota therefore applies over a short window, typically one to two years after the award.

PNRR contracts are awarded by many small, local contracting authorities. In our sample, XX distinct contracting authorities award at least one PNRR contract, against XX for ordi-

[18] We include contracts awarded through both competitive procedures and direct awards. Because we match contracts on the award procedure (Section 4.3), treated and control contracts are always awarded through the same type of procedure.

[19] Contract duration is observed for [XX] percent of contracts.

nary contracts. Municipalities award XX percent of PNRR contracts, regions and provinces XX percent, and central administrations XX percent. The median authority awards XX PNRR contracts over the period. Contracting authorities apply the full quota to most PNRR contracts. The full 30 percent quota applies to XX percent of PNRR contracts, a reduced quota to XX percent, and a waiver to XX percent. The most common reason for a waiver is [XX], cited in XX percent of waived contracts. Contracts with the full quota are [larger/smaller] than waived contracts, with a median value of €XX against €XX, and are more often awarded through [open/negotiated] procedures.

The firms that win PNRR contracts with the quota employ few women. Table **??** describes all firm-contracts in our sample in the baseline year, by type of contract won, and compares them with the average Italian private firm. The median winning firm employs XX workers, against XX in the private sector as a whole. Women account for XX percent of its employees and XX percent of its new hires, against XX and XX percent in the private sector. Only XX percent of winning firms already hire at least 30 percent women, so the quota binds for almost all of them.

Winning firms are frequent players in public procurement. Between 2017 and 2020, the median winning firm won XX public contracts, and XX percent of them won at least one contract every year. For these firms, the threat of exclusion from future calls is a real cost.

Hiring is common in winning firms, but rarely large. In a typical year before the award, XX percent of winning firms hire at least one worker and XX percent hire at least three, the scale at which the quota becomes operative in practice. The median firm hires XX workers per year, of whom XX are women. Female hires differ from male hires before the quota: XX percent of women are hired on temporary contracts, against XX percent of men; XX percent work part-time, against XX percent; and XX percent enter white-collar jobs, against XX percent. XX percent of female hires come from non-employment.

PNRR contracts are large relative to the size of the firms that win them. On average, the contract is worth 32 percent of the winning firm's annual sales, and the median contract 5.5 percent. The median contract is also worth XX percent of the winning firm's annual sales and XX times its annual wage bill. For XX percent of firms, the contract is worth more than XX percent of annual sales. A contract of this size is likely to require additional workers, which is when the quota applies.

4 Empirical Strategy

4.1 Comparing Winners of Similar Contracts

The main empirical challenge is selection into contracts with the quota. Contracting authorities decide which calls are covered by the quota before they publish them, and they can waive it for contracts that require few hires or specific skills. Firms observe the quota status before they bid. A simple comparison between firms that win contracts with and without the quota would therefore mix the effect of the quota with differences in the firms, sectors, contracting authorities, and projects involved, and in how many women these firms usually hire.

We address this problem by comparing firms that win nearly identical contracts, with and without the quota. Every firm in our comparison wins a public contract. Treated and control firms therefore face the same increase in demand, at the same time, from a contract with the same object and value. Each firm then decides whether to meet this demand with its existing workers or with new hires. The only difference between treated and control firms is that, for the treated firm, women must make up at least 30 percent of any new hires needed to execute the contract. Our estimates thus measure the effect of the quota, not the effect of winning a public contract.

4.2 Treatment and Comparison Groups

We define treatment at the firm level. For each firm, we keep only the first PNRR contract subject to the full 30 percent quota that it wins, and we call this award the treatment event. After the treatment event, a treated firm may win other public contracts, with or without the quota. This leaves 4,112 treated firm-contracts.

Control firm-contracts are contracts without any quota. They include ordinary contracts, which by definition carry no quota, and PNRR contracts for which the contracting authority waived the quota entirely. We exclude PNRR contracts with a reduced quota from both groups, since they impose a weaker version of the same requirement. Firms that ever win a contract with the full quota cannot serve as controls. A firm that wins contracts without the quota and later wins one with the full quota enters the treated group at the date of that later award, and its earlier contracts are not used as controls. The pool of potential controls contains 209,961 firm-contracts won by 41,481 firms.

The same control firm can appear more than once in our sample, for two reasons. First, a firm can win several contracts without the quota over time, and each of these firm-contracts is a separate potential control. Second, we match with replacement, so the same firm-

contract can serve as the control for more than one treated firm-contract. We account for this repetition when computing standard errors (Section 4.4).

4.3 Matching on Contracts and Firms

We match each treated firm-contract to one control firm-contract in two steps. The first step ensures that the two contracts are comparable. The second step ensures that the two firms were comparable before the award. Appendix Table A1 lists all criteria.

In the first step, we require the control contract to have the same object (works, services, or supplies) and the same award procedure as the treated contract, and to be awarded within six months of it. The two contracts must also share a CPV code. We search for a common code at the four-digit level first and move to three and then two digits only when no match exists.[20] Finally, the two contract values must differ by less than 10 log points and by less than €1,500 in absolute terms. Treated and control firms therefore win contracts of almost exactly the same value, for the same type of work, in the same period. Conditional on these characteristics, we argue that whether a contract carries the quota is as good as random from the point of view of the winning firm.

In the second step, we compare the firms that won these contracts. We require the control firm to operate in the same macro-sector as the treated firm, defined by the first digit of the ATECO classification. Its log employment in the baseline year must differ from that of the treated firm by less than 0.30, and its female employment share by less than 10 percentage points. Among the remaining candidates, we select the one closest to the treated firm in log baseline employment and baseline female share according to Jäger et al. (2024), using the standardized Euclidean distance.

The procedure yields 685 matched pairs, or 16.7 percent of treated firm-contracts. Each pair contains one treated firm-contract and its control, and the pair is the unit of comparison in our estimating equation: we compare the two firms in each pair with each other, before and after the award. The 685 controls correspond to 665 distinct firm-contracts, won by 642 distinct firms. Table 1shows that treated and control firms are nearly identical before the award. The average firm employs about 21 workers, of whom 2.6 are women, and wins a contract worth about €92,000. Female employment shares, white-collar shares, and weekly wages are also similar across the two groups.

[20] In the matched sample, 73 percent of pairs share a four-digit CPV code, 13 percent a three-digit code, and 14 percent a two-digit code.

Table 1: Baseline balance between all treated firms, matched treated firms, and matched controls

	All treated		Matched treated		Matched controls	
Variable	Mean	Median	Mean	Median	Mean	Median
Panel A: Variables used in matching						
Female employment share	0.11	0.10	0.11	0.11	0.11	0.11
Total employment (log)	3.10	2.89	2.63	2.48	2.63	2.48
Industry	0.13		0.13		0.13	
Services	0.16		0.13		0.13	
Construction	0.59		0.70		0.70	
Agriculture	0.00		0.00		0.00	
Multisector	0.12		0.04		0.04	
Procurement amount	836,793.47	140,667.52	92,790.81	47,638.07	92,799.21	47,607.00
Panel B: Main baseline outcomes						
Total employment	43.63	18.00	21.39	12.00	21.51	12.00
Female employment	5.39	2.00	2.63	1.00	2.60	1.00
Male employment	38.24	17.00	18.75	11.00	18.91	11.00
Blue-collar employment	25.17	12.00	13.73	9.00	14.32	9.00
White-collar employment	18.46	4.00	7.66	3.00	7.19	2.00
Weeks worked	1,684.72	653.00	839.24	463.00	834.05	468.00
Wage bill	1,207,789.97	367,138.50	556,132.69	251,897.00	526,392.38	252,418.00
White-collar employment share	0.32	0.22	0.27	0.20	0.25	0.18
Female white-collar employment share	0.10	0.09	0.11	0.10	0.10	0.10
Total hires	11.09	5.00	4.95	3.00	5.12	3.00
Female hires	1.08	0.00	0.44	0.00	0.45	0.00
Average weekly wage	576.79	524.18	545.12	508.44	546.84	510.56
Average weekly wage female	598.14	543.95	561.14	528.10	564.49	524.83
Average weekly wage (log)	6.25	6.23	6.21	6.20	6.22	6.20
Gender wage gap (%)	-1.44	-0.59	-1.94	-1.74	-2.79	-1.98
N. observations	4,112		685		685	
N. firm-events	4,112		685		665	
N. unique firms	4,112		685		642	

Notes: Statistics are measured in the baseline year. "All treated" reports eligible treated firm-events before matching; "Matched treated" and "Matched controls" report the matched firm-events used in the estimation. A control firm matched to different procurement events enters once for each event, possibly with different baseline years. Panel A reports the firm-level matching variables and Panel B selected baseline outcomes. The female share of white-collar employment is computed for firms with at least one white-collar employee, and the average female weekly wage for firms with at least one female employee. The gender wage gap is the log difference in firm-level FTE weekly wages between men and women, read as the percentage by which men's wages exceed women's, and is computed for firms with at least one worker of each gender.

Our matching criteria are deliberately strict, and we accept a smaller sample in exchange. Loose criteria would match many more treated firms, but to controls that win contracts of different value, object, or timing. Differences in outcomes would then reflect differences in the contracts won, not only the quota. With strict criteria, treated and control firms receive almost the same award, so the remaining difference between them is the quota itself. The cost is that we match only 16.7 percent of treated firm-contracts.

The matched sample therefore differs from the full set of treated firms in predictable ways. Appendix Table A2 shows that matched firms are smaller, with a median of 12 employees against 19, and win much smaller contracts, with a median of about €48,000 against €146,000. Large contracts are rarer, so they are harder to match on value within €1,500. Matched firms are also slightly more concentrated in construction (73 percent against 71 percent). The female employment share is the same in the two groups, at 11 percent. Our estimates therefore describe the response of small and medium firms, mostly in construction, to contracts of moderate size.

4.4 Estimating Equation

We estimate the effect of the quota with a difference-in-differences design within matched pairs. For firm i, which wins contract c in matched pair p, we observe outcome Y_{icpt} in calendar year t and estimate

$$Y_{icpt} = \beta \left(\text{Treated}_{icp} \times \text{Post}_{ict}\right) + \alpha_{ic} + \lambda_{pt} + \varepsilon_{icpt}. \tag{1}$$

Treated_{icp} equals one for the treated firm in pair p and zero for its control. Post_{ict} equals one in the years after firm i wins contract c. For control firms, Post_{ict} is defined by the award date of their own contract, which falls within six months of the treated award. Firm-by-contract fixed effects, α_{ic}, absorb all time-invariant differences between firm-contracts. Pair-by-year fixed effects, λ_{pt}, absorb all shocks common to the two firms in a pair in a given year, so each pair follows its own time path. The coefficient β compares the change in outcomes after the award for treated firms with the change for their matched controls. It measures the average effect of the quota on treated firms in the matched sample. Because the same control firm can appear more than once, we cluster standard errors at the firm level.

To study the timing of the effects and to test for pre-existing differences, we also estimate an event-study version of equation (1):

$$Y_{icpt} = \sum_{k \neq -1} \beta_k \left(\text{Treated}_{icp} \times \mathbf{1}\left[t - E_{ic} = k\right]\right) + \alpha_{ic} + \lambda_{pt} + \varepsilon_{icpt}, \tag{2}$$

where E_{ic} is the year in which firm i wins contract c, and k ranges from -3 to $+2$. The year before the award is the reference period. The coefficients β_k for $k < -1$ test whether treated and control firms followed similar paths before the award.

We estimate equations (1) and (2) by ordinary least squares for all outcomes; only the construction of the outcome changes. Some outcomes are in levels, such as the number of hires. Others are shares or rates, such as the hiring rate, defined as the number of hires divided by total employment in the year before the award. We describe each outcome when we present the results. To ease interpretation, we also report the estimates for hiring outcomes as a percentage of their baseline mean among treated firms.

4.5 Identifying Assumption and Threats

Our identifying assumption is that, without the quota, treated firms and their matched controls would have followed parallel paths after the award. Matching makes this assumption plausible: the two firms win contracts of the same value and type, at the same time, and had similar size and female employment shares before. The event-study estimates offer a direct test, and we find no evidence of differential trends before the award (Figure 2). Four concerns remain.

First, firms may select into calls with the quota on characteristics we do not observe. For example, firms that find it easier to recruit women may be more likely to bid for these calls. Matching on the baseline female share and the absence of pre-trends limit this concern. Selection should also be weak when the contract matters for the firm. When a contract is large relative to the firm's sales, giving it up to avoid the quota would be costly, and only a strong distaste for hiring women could justify it. We show in Section 5.6 that responses are concentrated precisely among these firms.[21]

Second, PNRR contracts with the full quota also require that 30 percent of new hires be younger than 36. The two requirements apply to different groups of workers and are likely to work through different mechanisms, so we focus on the gender requirement. Firms could meet both requirements at once by hiring young women.

Third, treated and control firms may operate in the same local labor markets. If treated firms hired women away from control firms, our estimates would overstate the effect of the quota. This concern is limited in our setting: about 75 percent of the additional women hired by treated firms come from non-employment rather than from other firms.

Fourth, we do not observe whether firms comply with the quota or whether penalties

[21] In future work, we plan to use data on all bidders, including runner-up firms. Comparing winners and runners-up in calls with and without the quota will allow us to separate the effect of the quota from the effect of winning a contract with a triple-difference design.

are applied (Section 2.2). We therefore estimate the effect of being exposed to the quota, not the effect of meeting the 30 percent target. Our estimates capture the quota as it was monitored and enforced in practice, which is the relevant parameter for policy.

5 How Firms Can Respond to a Hiring Quota

The quota aims to raise female employment, but its effect is not obvious ex ante. The quota applies only to the new hires needed to execute the contract, it sets only the share of women among them, and it lasts only as long as the contract. Every other choice is left to the firm, which keeps full discretion. A firm that wins a contract with the quota can therefore respond in several ways. In principle, more female hires do not necessarily translate into more female employment. Firms may let the new hires go shortly after hiring them, or replace women already employed in the firm with new ones, leaving the number of women unchanged. Some responses raise female employment, while others leave it unchanged or reduce hiring altogether. We organize these responses along three margins, each conditional on the previous one: whether the firm hires at all (the extensive margin), how many women and men it hires (the intensive margin), and which women it hires, for which jobs, and on what terms (the compositional margin). For each margin, we first describe how firms can respond and then present the evidence.

All these predictions are relative to the matched control firms. Treated and control firms win similar contracts and face the same increase in demand, so both may hire more after the award. When we say that the quota lowers the probability of hiring, we mean that treated firms hire less than their controls, not necessarily less than before the award. The same applies to every outcome: we estimate the differential effect of the quota, net of the effect of winning a contract.

These margins are not mutually exclusive. Different firms may respond in different ways, and the same firm may respond on several margins at once: it may hire less overall but more women among those it hires. The net effect of the quota on female employment is the sum of these responses. The rest of this section follows this order. We first show that the quota raises both female hiring, the outcome it targets directly, and female employment, the outcome it aims to change. We then decompose these effects: we estimate the effects on hiring and on the retention of new hires, which capture the extensive and intensive margins, and turn to the women hired and their jobs, which capture the compositional margin. Finally, we examine firm performance.

Figures report the event-study coefficients from equation (2), and the text reports the average post-award effect from equation (1). We describe the construction of each outcome

when we present it. Event time zero is the year of the award, and we observe firms from three years before to two years after it.

5.1 The Effect on Female Hiring and Female Employment

The quota raises the number of women that treated firms hire. After the award, treated firms hire 0.11 more women per year than their controls, an increase of 25.9 percent over a baseline of 0.44 (Figure 2, Panel A).[22] The increase is concentrated in the year of the award and the following year, when the contract is being executed and the quota applies. By the second year after the award, female hiring returns close to its pre-award level.

We want to measure the effect on employment in percentage terms, as we would with logs. Logs, however, are not defined for firms with no female employees. We therefore build the outcome in two steps. First, we divide the number of women employed by the firm in each year by its total employment in the year before the award. This puts firms of different size on the same scale and keeps firms with no women. Second, we divide this ratio by the average female share of employment among treated firms in the year before the award. The outcome therefore measures female employment relative to that of the average treated firm, and its changes can be read as percentage changes. Consider a firm with 20 employees, 2 of them women, when the average female share is 10 percent. Its outcome equals $(2/20)/0.10 = 1$. If the firm employs one more woman, the outcome rises to $(3/20)/0.10 = 1.5$, an increase of 50 percent. We build male employment in the same way, using the average male share.

These percentage changes are measured relative to the average treated firm, not to each firm's own baseline. The two coincide when firms have similar female shares before the award. This is a reasonable approximation in our sample, where female shares are low in almost all firms.

The increase in female hiring carries over to employment. Female employment rises by 11.2 percent after the award, while male employment does not change (Figure 2, Panel B). Before the award, female employment in treated and control firms follows the same path. It starts to rise in the year of the award and keeps growing in the following year. Male employment remains flat throughout. Because female and male employment are divided by the same baseline employment, comparing the two coefficients also tells us how the female share of employment changes: the share rises when the coefficient for women exceeds the one for men (Appendix B). With a coefficient close to zero for men, the female share of employment in treated firms rises.

[22] The quota sets the share of women among new hires, but we cannot use this share as an outcome: it is not defined in the years in which a firm hires no one. We therefore study the number of female hires, and the number of male hires in Section 5.3.

The quota therefore changes whom firms hire, and this change reaches the stock of employment without displacing men. The two percentages are not directly comparable: hires are a flow measured against a small baseline, while employment is a stock. The increase in female employment also occurs even though, as we show below, some treated firms hire less than their controls. The rest of this section shows how firms produce these effects.

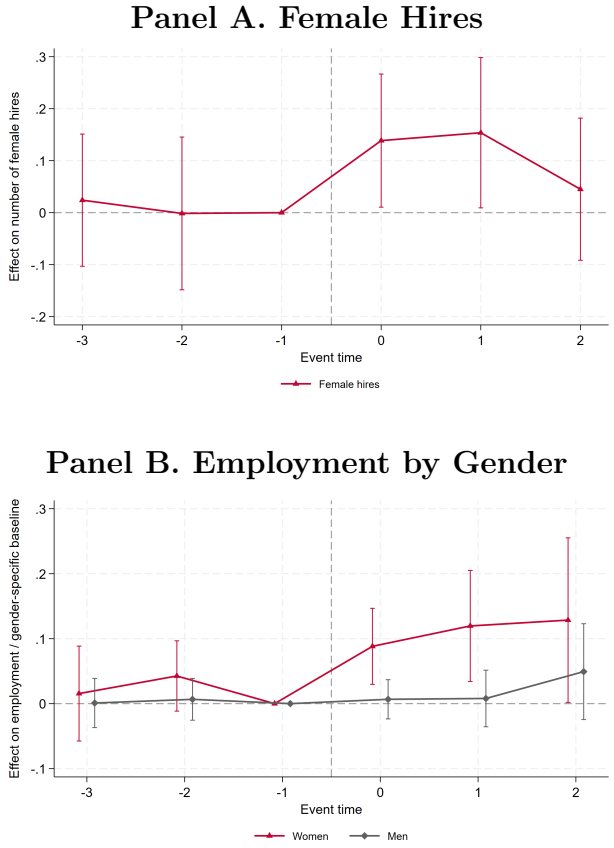


Figure 2: Female Hires and Employment by Gender

Notes: The figure reports event-study coefficients from equation (2), comparing treated firms with their matched controls. Event time zero is the year of the award; the year before the award is the reference period. Vertical bars show 95 percent confidence intervals, based on standard errors clustered at the firm level. In Panel A, the outcome is the number of women hired in the year. In Panel B, the outcome for gender g is $Y_{it}^{g} = \left(N_{it}^{g}/N_{i,-1}\right)/\overline{s}_{-1}^{g}$, where N_{it}^{g} is the number of employees of gender g in firm i in year t, $N_{i,-1}$ is the firm's total employment in the year before the award, and $\overline{s}_{-1}^{g}$ is the average share of gender g in employment among treated firms in the year before the award. Coefficients can be read as percentage changes relative to the average treated firm.

5.2 Extensive Margin: Whether to Hire

The first decision is whether to hire at all. Once the contract is awarded, the firm must comply with the quota only if it hires at least three workers for the project. A firm can therefore avoid the constraint by recruiting fewer than three people. It can execute the

project with the workers it already employs, moving them from other activities or asking them to work more hours. It can also strategically time its recruitment around the contract, hiring before the award, when the quota does not yet apply.

Escaping the quota is costly, and the cost grows with the importance of the contract for the firm. Avoiding the quota is harder when the contract is large: workers must be reassigned from other tasks, and execution may slow down, with possible delays and penalties (Section 2.1). Ignoring the quota is risky: a firm that hires three or more workers without meeting it faces penalties, the termination of the contract, and exclusion from future calls. How large this risk is depends on how firms expect the quota to be enforced, which is uncertain (Section 2.2). When the contract accounts for a large share of the firm's activity, even a small chance of losing it would be costly and hard to justify. Contract importance therefore works as a compliance cost, and we expect firms to comply more when the contract matters more. We return to this prediction in Section 5.6.

Avoiding and complying with the quota have opposite implications for hiring. If firms avoid the quota, the probability that they hire at least three workers should fall after the award. Firms that stay just below the threshold may still hire one or two workers, so this decline can be larger than the decline in the probability of any hire. If firms avoid only the recruitment that the quota would constrain, the decline should be concentrated in hires of men. If firms comply instead, they may hire women they would not otherwise have hired, raising the probability of hiring at least one woman and possibly the number of hires overall. We measure this margin with the probability that the firm hires at least one worker, overall and by gender.

Some firms respond to the quota by hiring less. The probability that a firm makes any hire falls by 2.8 percentage points from a baseline of 87 percent. Figure 3 shows that this decline comes entirely from men. The probability of hiring at least one man falls by 3.0 percentage points, from a baseline of 86 percent, and the drop is largest in the year of the award. The probability of hiring at least one woman does not change. This pattern is consistent with firms executing the contract with their existing workforce rather than recruiting under the quota. Because we do not observe which workers are assigned to each contract, we cannot test this channel directly.

5.3 Intensive Margin: How Many Women and Men to Hire

Firms that recruit must decide how many women and men to hire. Two responses meet the quota. In the first, firms expand: they hire women on top of the men they would have hired anyway, so total hires rise. In the second, firms substitute: they hire women instead of some

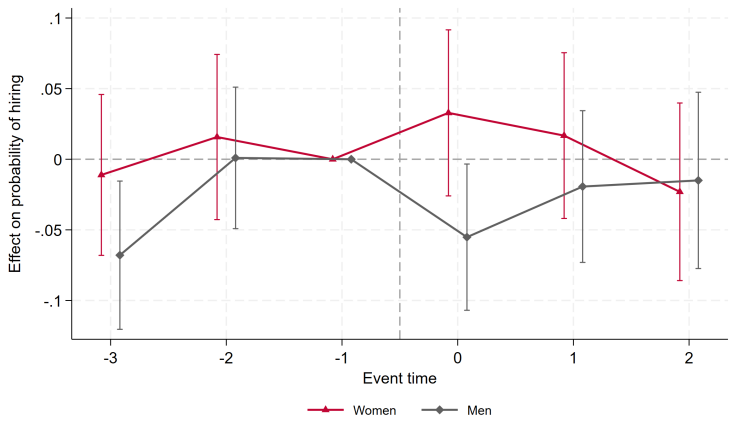


Figure 3: Probability of Hiring, by Gender

Notes: The figure reports event-study coefficients from equation (2), comparing treated firms with their matched controls. Event time zero is the year of the award; the year before the award is the reference period. Vertical bars show 95 percent confidence intervals, based on standard errors clustered at the firm level. The outcome is an indicator for hiring at least one worker of each gender in the year.

of the men they would have hired, so the female share of new hires rises while total hires do not change. In principle, firms could also reach the required share by hiring fewer men without hiring more women, but this would mean giving up workers needed to execute the contract.

Changes in hires do not translate mechanically into changes in employment. New female hires raise female employment only if they stay in the firm and do not replace women already employed there. The size of the existing workforce also matters. Because the quota applies to the flow of new hires, its effect on the stock of employment depends on how many workers the firm hires relative to how many it already employs. In a small firm, three new hires can change the gender composition of the workforce. In a large firm that hires few workers a year, even full compliance would barely move the female share of employment. The same logic applies to men: hiring fewer men does not necessarily reduce male employment when men make up most of a large workforce. For this reason, we scale hires by baseline employment and study heterogeneity by firm size (Section 5.6). We measure this margin with the number of female and male hires and with the female hiring rate.

Firms that recruit hire more women, and the effect is large relative to the hiring these firms do. The average increase of 0.11 women per firm and year is small partly because many firms hire no women at all. Scaled by the size of the firm, the response is large. The female hiring rate, defined as female hires over baseline employment, rises by 0.7 percentage points, or 35.8 percent of its baseline value of 2.0 percent (Figure 4).

These hiring choices add up to the increase in female employment documented in Sec-

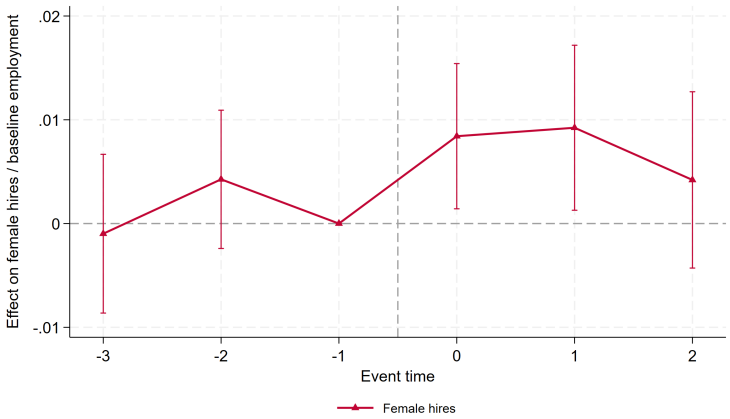


Figure 4: Female Hiring Rate

Notes: The figure reports event-study coefficients from equation (2), comparing treated firms with their matched controls. Event time zero is the year of the award; the year before the award is the reference period. Vertical bars show 95 percent confidence intervals, based on standard errors clustered at the firm level. The outcome is the number of women hired in the year divided by baseline employment.

tion 5.1, with no change in male employment.

Whether new female hires stay in the firm is therefore an outcome in its own right. Firms that hire women only to meet the quota may keep them only as long as the quota applies, and let them go once the contract ends. Firms that find the new hires productive, or that face high costs of hiring and training, may instead keep them. We study separations and the retention of new female hires both while the quota applies and after it expires. Comparing the two periods tells us whether the employment gains last beyond the contract.

The employment gains show signs of fading once the quota expires. Female separations do not increase in the year of the award or in the following year (Figure 5). Two years after the award, when most contracts have ended and the quota no longer applies, separations rise by about 0.15 women per firm. Net female hires, defined as female hires minus female separations, follow the same pattern: they rise by about 0.2 in the year of the award and fall back to zero or below two years later. The number of women hired in the year of the award who are still employed by the firm two years later, divided by baseline employment, also rises. We find no such effect for women hired one year after the award. Female employment remains above its pre-award level two years after the award, but the estimate is imprecise. These patterns are consistent with the large share of temporary contracts we document in Section 5.4: many of the additional jobs end with the contract that created them.

These results show that firms adjust on the extensive and intensive margins at once. Some firms avoid the quota by not recruiting, and this response falls on men. Firms that recruit hire more women, and this response raises female employment. The second effect

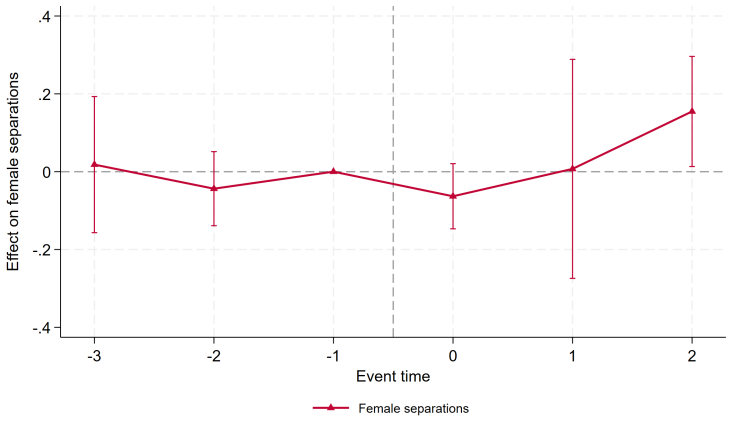


Figure 5: Female Separations

Notes: The figure reports event-study coefficients from equation (2), comparing treated firms with their matched controls. Event time zero is the year of the award; the year before the award is the reference period. Vertical bars show 95 percent confidence intervals, based on standard errors clustered at the firm level. The outcome is the number of women employed by the firm in the previous year who are no longer employed there in the current year.

dominates: on net, the quota increases the number of women employed in treated firms.

5.4 Compositional Margin: Which Women, in Which Jobs, and on What Terms

Firms that hire women must also decide whom to hire and where to place them. The quota does not say where these women should come from: firms can hire women who were not employed or women already working in other firms. Nor does it say which jobs women should fill. Where production relies on physically demanding tasks, as in construction, firms may meet the quota by hiring women into administrative and technical roles, leaving the composition of the production workforce unchanged. Construction makes up most of our sample, and it is the sector where differences in productivity between men and women in production tasks are most plausible. This makes construction a demanding test: if the quota works here, it is likely to work where women are easier to hire (Section 2.4). We measure these choices with the previous labor market status and the occupation of new female hires. A new hire comes from another firm if the worker had a contract with a different private employer in the previous year, and from non-employment otherwise (Section 3.1).

Most of the additional women hired under the quota come from non-employment. Figure 6 decomposes the 35.8 percent increase in the female hiring rate by the previous labor market status of the women hired. Hires from non-employment account for 26.8 percentage points of the increase, and hires from other firms for 9.0 points. About 75 percent of the

additional female hires were therefore not employed before. The quota does not simply move women from one employer to another; it draws new women into employment.

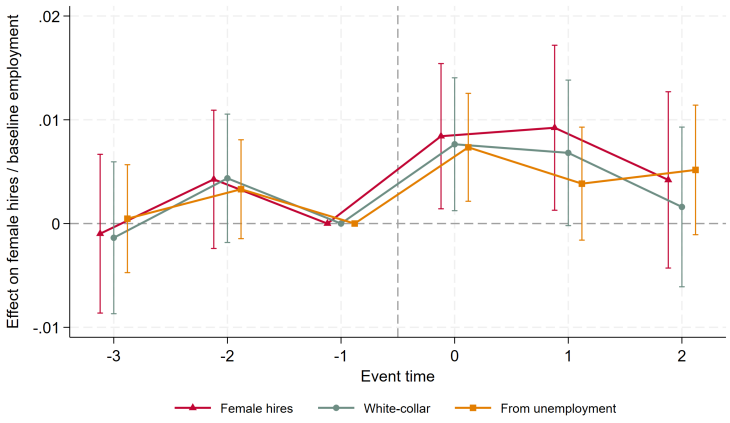


Figure 6: Composition of Additional Female Hires

Notes: The figure decomposes the average post-award effect on the female hiring rate (35.8 percent of a baseline of 2.0 percent) by the previous labor market status of the women hired (top bar) and by their occupation (bottom bar). Each segment reports the effect on the corresponding hiring rate, as a percentage of the baseline female hiring rate.

The additional women are mostly hired into white-collar jobs. White-collar hires account for 26.3 percentage points of the increase in the female hiring rate, and blue-collar hires for 9.5 points, so 74 percent of the additional women enter white-collar positions. This is the pattern we expected in a sample dominated by construction: firms meet the quota mainly in clerical, administrative, and technical roles rather than in production jobs.

The same freedom applies to the terms of employment. Because the quota does not regulate wages, hours, or contract types, firms can meet it at low cost: with temporary contracts that end with the project, with part-time jobs, or with lower wages. Alternatively, if hiring is costly, firms may offer the women they recruit the same terms as their regular workforce. We measure the terms of employment with contract type, working time, and weekly wages.

The jobs offered to the additional women are mostly temporary and part-time. Figure 7 decomposes the increase by contract type and working time. Temporary contracts account for 29.1 percentage points of the 35.8 percent increase, or 81 percent of the additional hires. Part-time contracts account for 22.8 points, or 64 percent. Firms use the freedom the quota leaves them to offer short and flexible contracts.

Wages do not change. Figure 8 shows that the weekly wages of women and men in treated firms follow the same path as in control firms, before and after the award. The estimates are close to zero. We find the same result for new hires and for white-collar workers. Firms do

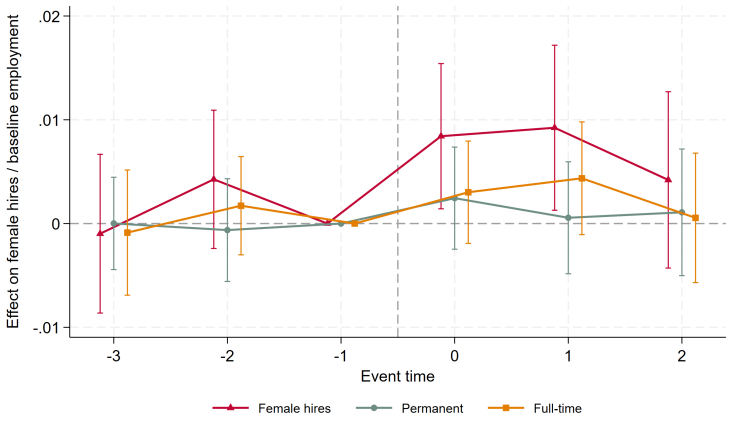


Figure 7: Terms of Employment of Additional Female Hires

Notes: The figure decomposes the average post-award effect on the female hiring rate (35.8 percent of a baseline of 2.0 percent) by contract type (top bar) and working time (bottom bar). Each segment reports the effect on the corresponding hiring rate, as a percentage of the baseline female hiring rate.

not pay women less to comply with the quota, and they do not raise wages to attract them.

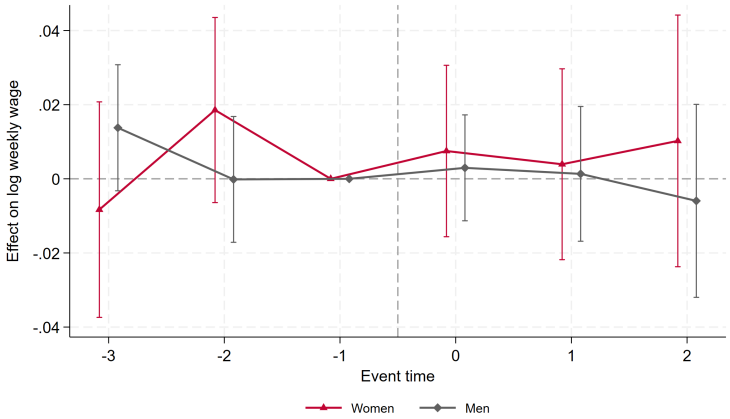


Figure 8: Weekly Wages by Gender

Notes: The figure reports event-study coefficients from equation (2), comparing treated firms with their matched controls. Event time zero is the year of the award; the year before the award is the reference period. Vertical bars show 95 percent confidence intervals, based on standard errors clustered at the firm level. The outcome is the log weekly wage of female and male employees.

Together, these results describe both the reach and the limits of the quota. The quota brings women who were not employed into jobs, and it places them mostly in white-collar positions. But these jobs are mostly temporary and part-time, and they pay the same wages.

5.5 Firm Performance

These responses can also affect how firms perform. If the quota forces firms to hire workers they would not otherwise choose, or to spend more on recruitment, it may lower productivity and profits. If instead it pushes firms to search in a wider pool of workers, it may improve the match between workers and jobs and leave performance unchanged or higher. We measure firm performance with labor productivity, revenues, total assets, and profit margins.

The quota does not harm firm performance. Figure 9 reports the difference-in-differences estimates for four outcomes from Orbis balance sheets. Log labor productivity rises by 6.7 percent after the award, and the effect is statistically significant. The estimates for log revenues, log total assets, and the profit margin are positive but small and not statistically different from zero, with point estimates of about 5, 3, and 5 percent.

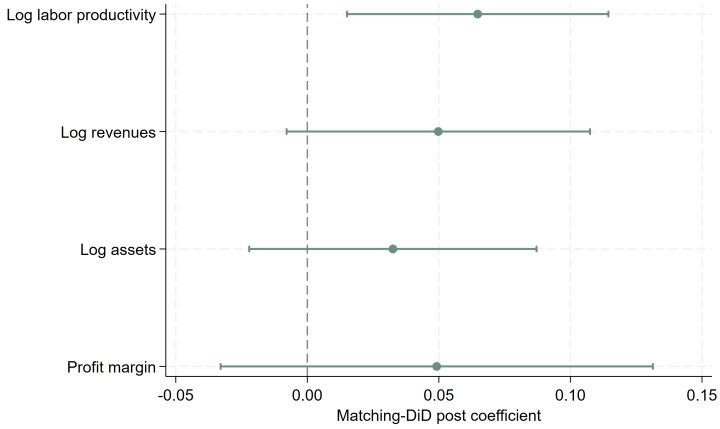


Figure 9: Firm Performance

Notes: The figure reports the average post-award effects from equation (1) for outcomes from Orbis balance sheets. Horizontal bars show 95 percent confidence intervals, based on standard errors clustered at the firm level.

We interpret the productivity result with caution. Labor productivity is value added per employee, as reported in Orbis, so it depends on both output and the number of employees, and many of the additional women work part-time. The confidence interval is also wide, ranging from about 1.5 to 11 percent. What the evidence shows clearly is that firms subject to the quota do not perform worse than comparable firms. If the quota imposed large costs on firms, through worse matches or higher hiring costs, we would expect lower productivity or profits. We find neither.

5.6 Heterogeneity

Responses are concentrated where the contract matters most for the firm. We measure contract importance as the value of the contract relative to the treated firm's average annual wage bill in the two years before the award. We split pairs into three groups using the terciles of this ratio among treated firms, and assign each control firm to the group of its matched treated firm. Figure 10 compares the bottom and top terciles. For contracts in the top tercile, the probability of any hire falls by 20 percent of its baseline, and the female share of employment, defined as the number of women employed in the year divided by total employment in the same year, rises by 15 percent of its baseline. For contracts in the bottom tercile, neither effect is distinguishable from zero. When the contract is large relative to the firm, the firm cannot easily ignore the quota, and it adjusts on both margins. This result also limits concerns about selection: firms for which the contract matters most have the least room to avoid it (Section 4.5).

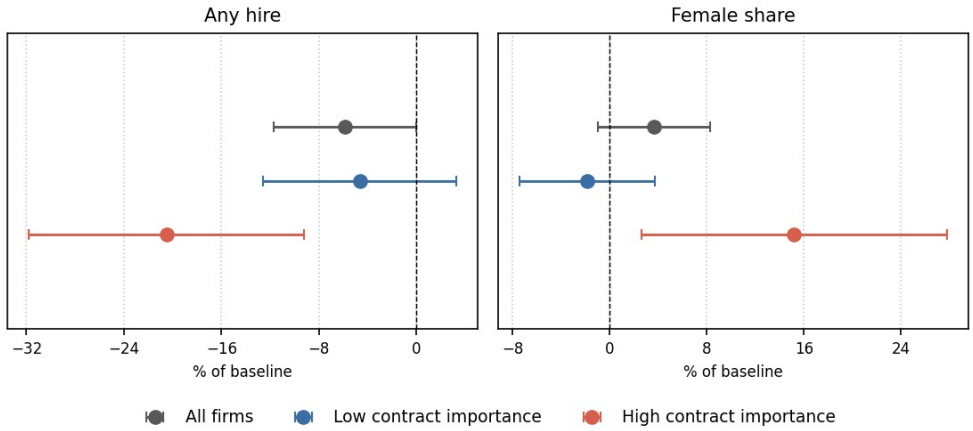


Figure 10: Heterogeneity by Contract Importance

Notes: The figure reports the average post-award effects from equation (1), as a percentage of the baseline mean, for all firms and separately for the bottom and top terciles of contract importance, defined as the value of the contract divided by the treated firm's average annual wage bill in the two years before the award. Terciles are computed among treated firms, and each control firm is assigned to the group of its matched treated firm. Horizontal bars show 95 percent confidence intervals.

Small firms drive the employment gains, while hiring avoidance is concentrated among mid-sized firms. Figure 11 splits firms into three groups by employment at baseline, using the terciles of the distribution of baseline employment among treated firms: small (up to 9 employees), middle (10 to 17), and large (more than 17). Each control firm is assigned to the group of its matched treated firm, so that treated and control firms are always compared within the same group. In small firms, the female share of weeks worked rises by about 13 percent, female hiring rises by about 85 percent, and female hiring from non-employment

nearly doubles. In mid-sized firms, the probability of any hire falls by about 10 percent, with no gain in female employment. Large firms show no significant response on any margin. For a small firm, a single new hire can be enough to meet the quota, and the contract is more likely to be large relative to the firm. Mid-sized firms may instead have enough workers to staff the contract internally, which makes avoidance easier.

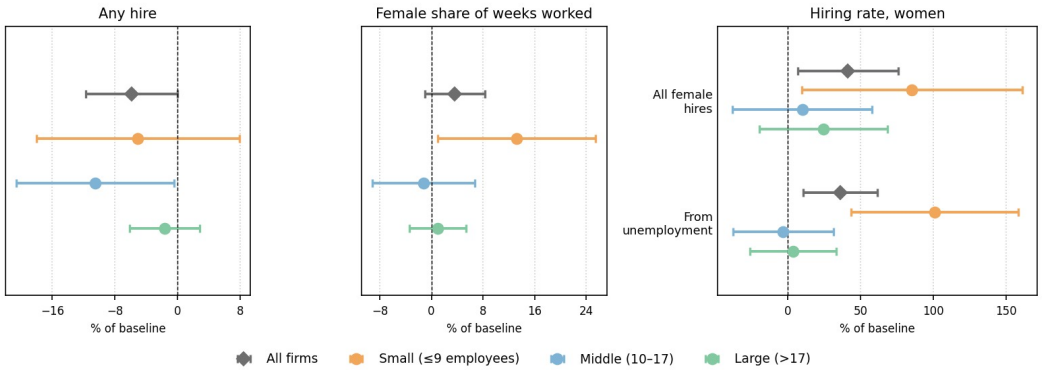


Figure 11: Heterogeneity by Firm Size

Notes: The figure reports the average post-award effects from equation (1), as a percentage of the baseline mean, for all firms and separately for small (up to 9 employees), middle (10 to 17), and large (more than 17) firms. The groups correspond to the terciles of baseline employment among treated firms; each control firm is assigned to the group of its matched treated firm. Horizontal bars show 95 percent confidence intervals.

6 Conclusions

We study a gender quota on new hires attached to public contracts in Italy. Firms that win contracts with the quota employ more women than comparable firms that win similar contracts without it, with no change in male employment and no evidence of efficiency losses. Most of the additional women come from non-employment and enter white-collar jobs, but these jobs are mostly temporary and part-time, and part of the gain fades once the contract ends. Some firms avoid the requirement by hiring less.

Three lessons follow. First, binding conditions attached to public contracts can change the hiring of private employers, and their reach grows with firms' reliance on public demand. Second, the effects of a temporary quota last longer when replacing workers is costly, so hiring quotas may have more persistent effects in rigid labor markets, such as Italy's, than in labor markets with employment at will. Third, even when the jobs created by the quota end, they may strengthen women's attachment to the labor market. Studying the later careers of the women hired under the quota is a natural next step.

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A Matching and Sample

B Employment Outcome and the Female Share

This appendix shows why comparing the coefficients for women and men tells us how the female share of employment changes.

Table A1: Matching Criteria

Criterion	Rule
Contracts	
Contract object	Same object: works, services, or supplies
Award procedure	Same procedure, including whether the contract is competitive or directly awarded
Timing	Control contract awarded within six months of the treated contract
Product classification	Same CPV code, searched at four digits, then three, then two
Contract value	Difference in log value below 0.10
Firms	
Sector	Same macro-sector (first digit of the ATECO classification)
Size	Difference in log employment in the baseline year below 0.30
Female share	Difference in the female share of employment in the baseline year below 10 percentage points
Selection among candidates	Nearest neighbor in standardized Euclidean distance, with replacement

Table A2: Matched and Unmatched Treated Firms

	Matched	All treated
Employment (median)	12	19
Female share of employment	0.11	0.11
Contract value (median, euros)	48,000	146,000
Construction	0.73	0.71
Firm-contracts	685	4,112

Notes: The table compares treated firm-contracts that enter the matched sample with all treated firm-contracts. Characteristics are measured in the year before the award.

For firm i, let F_{it} and M_{it} be female and male employment in year t, and $N_{i,-1}$ total employment in the year before the award. The outcomes are

$$Y_{it}^F = \frac{F_{it}}{N_{i,-1}\,\overline{s}_{-1}^F}, \qquad Y_{it}^M = \frac{M_{it}}{N_{i,-1}\,\overline{s}_{-1}^M},$$

where $\overline{s}_{-1}^F$ and $\overline{s}_{-1}^M = 1 - \overline{s}_{-1}^F$ are the average female and male shares among treated firms in the year before the award. A treatment effect β^F on Y^F corresponds to a change in female employment of $\Delta F = \beta^F\,\overline{s}_{-1}^F N_{i,-1}$, and similarly $\Delta M = \beta^M\,\overline{s}_{-1}^M N_{i,-1}$.

The female share is $s = F/(F+M)$. For small changes,

$$\Delta s \approx \frac{M\,\Delta F - F\,\Delta M}{(F+M)^2}.$$

For a firm with the average baseline shares, $F = \overline{s}_{-1}^F N_{i,-1}$ and $M = \overline{s}_{-1}^M N_{i,-1}$. Substituting,

$$\Delta s \approx \overline{s}_{-1}^F\,\overline{s}_{-1}^M\left(\beta^F - \beta^M\right).$$

The female share therefore rises if and only if $\beta^F > \beta^M$. The same argument applies to any two groups that together make up the workforce, such as white-collar and blue-collar workers.